

A New Analysis of the Morphological Structure of Japanese Verbal Adjectives

Shinya Makino

itakanro@g.ecc.u-tokyo.ac.jp

Abstract:

This paper reanalyses the morphological structure of Japanese verbal adjectives — so-called *i*-adjectives. Traditionally, inflected forms of verbal adjectives, such as *taka-ku* from the verbal adverb *takai* "to be high", have been analysed as consisting of the stem *taka-* and the inflectional suffix *-ku*. However, this analysis has limitations. Considering the syntactic, semantic, and morphological parallelism with verb stems — such as *tabe-* from the verb *taberu* "to eat" — it can be inferred that *taka-ku* as a whole corresponds to the verbal adjective stem. This paper analyses *taka-* as one of the two immediate constituents of the VA stem *taka-ku*, which represents the core lexical meaning of the verbal adjective as a VA radical, and *-ku* as the other constituent, functioning as the VA thematic suffix that licences the VA radical as a syntactic constituent within the VA category at the sentential level. The paper then examines how verbal adjectives encode the [–past] tense proposing a new understanding based on historical and morphological perspectives. Rather than the traditional interpretation that *-i* expresses non-past, it is argued that the phonologically zero inflectional suffix *-0* denotes [–past] in opposition to *-ta* specified as [+past]. Based on this, the paper further explores the allomorphy of the VA thematic suffixes *-i* $\approx$ *-(k)i* $\approx$ *-ku* $\approx$ *-ke* and investigates how their distribution is determined by syntactic environments. Finally, the paper explores the similarities between the morphological structures of verbal adjectives and true verbs — particularly vowel-stem verbs.

1. The stem of verbal adjectives — *i*-adjectives

Traditionally, the sentence-final form of verbal adjectives ending in *-i* — that is, *i*-adjective — has been analysed as consisting of a stem followed by the inflectional ending *-i*. For instance, in *taka-i* "to be tall", *taka-* is considered the stem, and *-i* the ending. On the other hand, the non-past form of verbs — particularly vowel-stem verbs — has been analysed as consisting of a stem followed by the inflectional ending *-ru*. For instance, in *tabe-ru* "to eat", *tabe-* is considered the stem, and *-ru* the ending. **Although *taka-* and *tabe-* are both equally regarded as stems, their syntactic behaviour differs significantly.** The verbal stem *tabe-* without overt ending can appear on its own within a sentence as in (1), whereas the so-called adjectival stem *taka-* can never occur independently in such contexts as shown in (2):

- (1) Kare wa gohan o **tabe**, sosite deteiku / deteita. "Having a meal, he then leaves/left."
- (2) *Kanozyo wa se ga **taka**, sosite utsukusii / utsukusikatta. (ungrammatical)
"Being tall, she is / was beautiful."

What can occur in the same syntactic context as the verbal stem *tabe-* is not the so-called adjectival stem *taka-* itself, but rather the conjunctive form *taka-ku*, which has traditionally been analysed — like *taka-i* — as consisting of the so-called stem *taka-* and the so-called inflectional ending *-ku*, as in (3):

- (3) Kanozyo wa se ga **taka-ku**, sosite utsukusii / utsukusikatta. (grammatical)
"Being tall, she is / was beautiful."

As is clear from (1) and (3), at the stage of *Kare wa gohan o tabe* and *Kanozyo wa se ga taka-ku*, it remains indeterminate whether the utterance refers to a past or non-past situation. Temporal specification is only achieved by the final predicate — *deteiku* / *utsukusii* in the non-past, and *deteitta* / *utsukusikatta* in the past. In this respect as well, the verbal stem *tabe-* and the form *taka-ku* share the property of being **atemporal**: they do not themselves denote temporal reference. Furthermore, the similarity between the verbal stem *tabe-* and the form *taka-ku* is not limited to this point; as shown in (4) - (7), both can appear in the sentence followed by the suffixes *-te*, *-temo* and *-tatte*:

- (4) Kare wa gohan o tabe-te, deteita. "Having a meal, he then left."
- (5) Kanozyo wa se ga taka-ku-te, utukusikatta. "Being tall, she was beautiful."
- (6) Kare wa gohan o tabe-temo / tabe-tatte, hutoranai.
"Even having a meal / Even if he eats, he does not gain weight."

(7) Kanozyo wa se ga taka-ku-temo / taka-ku-tatte, medatanai.

"Even being tall / Even if she is tall, she does not stand out."

In this way, **there is clear syntactic, semantic, and morphological parallelism between *tabe-* and *taka-ku*. Given that *tabe-* is analysed as the stem of the verb, it seems more appropriate to treat *taka-ku* — not as the combination of the so-called stem *taka-* and the so-called inflectional ending *-ku*, as traditionally assumed — but rather as the stem of the verbal adjective in its entirety, which we call henceforth VA stem.** This raises the following question: how, then, should *taka-*, traditionally regarded as the stem, and *-ku*, traditionally regarded as the inflectional ending, be reanalysed morphologically as constituent morphemes? This issue will be addressed in the following section.

2. The radical and the thematic suffix of verbal adjectives

When we observe the inflected forms of the verbal adjective *taka-i* — namely, *taka-(s)i* $\approx$ *taka-(k)i* $\approx$ *taka-ku* $\approx$ *taka-ku-te* $\approx$ *taka-ku-temo* $\approx$ *taka-ku-tatte* $\approx$ *taka-k-at-ta* $\approx$ *taka-k-at-tar-ō* $\approx$ *taka-k-at-tar-a(ba)* $\approx$ *taka-k-at-tar-i* $\approx$ *taka-k-ar-ō* $\approx$ *taka-k-ar-e* $\approx$ *taka-ke-reba* $\approx$ *taka-ke-redo(mo)* — we find that they all share the element *taka-*. This *taka-* is also attested in derived nouns such as *taka-sa* "height" and *taka-mi* "loftiness", as well as in compound expressions such as *taka-nozomi* "high hope" and *taka-dai* "high platform", where it refers to the very property or state of "being tall". Since it cannot be further decomposed into smaller morphemes, it is to be regarded as a root for the moment. As the ungrammaticality of (8) clearly shows, however, the root *taka-* alone cannot function as an adjective — that is, as a category A — capable of establishing syntactic relations with other constituents within the sentence:

(8) *Koko kara taka sobieru tō ga mieru. (ungrammatical)

"From here, a tower rising high can be seen."

By contrast, example (9), in which *taka-* is accompanied by *-ku*, is grammatical:

(9) Koko kara taka-ku sobieru tō ga mieru. (grammatical)

"From here, a tower rising high can be seen."

Now, as the derived words and compound words cited above show, the root *taka-* functions only as a mere constituent of word formation: it cannot itself serve as a syntactic unit at the sentence level. We are thus led to posit that *taka-* requires an additional element in order to function as a

syntactic category A at the sentence level. This element is *-ku*, which is present in the sentential form *taka-ku*, but absent from word-level forms such as *taka-sa*, *taka-mi*, *taka-nozom-i*, and *taka-dai*. As observed in Section 1, the addition of *-ku* to the root *taka-* yields a verbal adjective stem functionally — syntactically, semantically, and morphologically — equivalent to the verbal stem *tabe-*. We shall therefore refer to *-ku* as the **VA thematic suffix**.

On the other hand, *taka-* was previously referred to as a root in relation to *-ku*, but this is not strictly accurate. As *na-daka-* in the compound verbal adjective *na-daka-i* "to be famous" in (10) shows, what precedes the VA thematic suffix need not be a single root, but may also involve root–root compounding:

(10) Yamazaki wa **na-daka-i** gengogakusya da. "Yamazaki is a famous linguist."

It is therefore morphologically inappropriate to refer to the constituent preceding the VA thematic suffix *-ku* as a root. In what follows, we shall refer to this element as a **VA radical**, in the sense of "one of the two immediate constituents of a verbal adjective stem that constitutes its semantic core".

As the inflected forms *taka-ku*, *taka-ku-te*, *taka-ku-temo* and *taka-ku-tatte* show, the verbal adjective stem formed with the suffix *-ku* — namely, *taka-ku* — can appear on its own, or followed by the suffixes *-te*, *-temo* and *-tatte*. From this, the morphological structure of verbal adjectives may be analysed as in (11):

(11)

VA stem		Ending
VA radical	VA thematic suffix	Inflexional suffix
<i>taka-</i>	<i>-ku</i>	—
<i>taka-</i>	<i>-ku</i>	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	<i>-tatte</i>

On the other hand, in the inflected forms *taka-k-at-ta*, *taka-k-at-tar-ō*, *taka-k-at-tar-a(ba)*, *taka-k-at-tar-i*, *taka-k-ar-ō* and *taka-k-ar-e*, if the focus particle *mo* is inserted before the conjugated forms of the light verb *ar-u* — namely, *at-ta*, *at-tar-ō*, *at-tar-a(ba)*, *at-tar-i*, *ar-ō* and *ar-e* — and the string is thereby divided into two words, the following alternations in (12) can be observed:

(12) *taka-k-at-ta* $\approx$ *taka-ku#mo#at-ta*
taka-k-at-tar-ō $\approx$ *taka-ku#mo#at-tar-ō*
taka-k-at-tar-a(ba) $\approx$ *taka-ku#mo#at-tar-a(ba)*
taka-k-at-tar-i $\approx$ *taka-ku#mo#at-tar-i*
taka-k-ar-ō $\approx$ *taka-ku#mo#ar-ō*

taka-**k**-ar-e $\approx$ taka-**ku**#mo#ar-e

From this, it can be seen that the VA thematic suffix *-ku* undergoes vowel deletion and appears in the form *-k-* when it precedes the stem of the light verb *ar-u*, namely *ar- $\approx$ at-*.

The reason why the inflected forms in (12) contain *ar-* or *at-*, the stem of the light verb *ar-u*, is that the endings *-ta*, *-tar-ō*, *-tar-a(ba)*, *-tar-i*, *-ō* and *-e*, which express tense and modality, combine only with verbal stems. In other words, *taka-ku* is a non-verbal stem and thus cannot combine directly with these endings. Consequently, the verbal adjective stem *taka-ku* combines with the stem *ar-* of the light verb *ar-u*, forming a compound verbal stem *taka-k-ar-* (= *taka-ku* + *ar-*), which as a whole possesses verbal status. This is made possible by the fact that Japanese is a head-final language, whereby the light verb stem *ar-* determines the category of the entire compound. Under this interpretation, the structure of (11) may be more accurately represented as in (13):

(13)

(Complex) stem			Ending
VA stem		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-ku</i>	—	—
<i>taka-</i>	<i>-ku</i>	—	<i>-te</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-ta</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-tar-a(ba)</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-k</i>	<i>ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-k</i>	<i>ar-</i>	<i>-e</i>

Besides the both morphologically and phonologically conditioned allomorphy *-ku $\approx$ -k*, the VA thematic suffix exhibits another type of allomorphy *-ku $\approx$ -i $\approx$ -(k)i $\approx$ -ke* depending on the syntactic environment in which it appears, as shown in (14) - (19):

(14) *Kanozyo wa se ga taka.

(15) Kanozyo wa se ga taka-i. "She is tall."

(16) *Asoko ni se no taka otoko ga iru.

(17) Asoko ni se no taka-i otoko ga iru. (Classical: taka-ki) "There is a tall man over there."

(18) *Se wa taka-reba ii to yū wake de wa nai.

(19) Se wa taka-ke-reba ii to yū wake de wa nai.

"It is not always the case that it is good if the height is tall."

Among these, *taka-ke* in (18), which bears the allomorph *-ke*, occurs in modern Japanese

exclusively before the suffix *-reba*, which marks a conditional meaning with affirmative polarity. By contrast, the syntactic conditions that determine the occurrence of *taka-i* in (15) and *taka-(k)i* in (17), bearing the allomorphs *-i* and *-(k)i* respectively, will be examined in the following section.

3. Syntactic conditions governing the distribution of the allomorphs *-i* and *-(k)i*

As observed in (3), the conjunctive form *taka-ku*, in which the VA thematic suffix appears in the form *-ku* and no inflectional ending follows, is atemporal. The fact that the form *taka-ku* lacks temporal reference is consistent with the interpretation that it does not bear any inflectional suffix expressing tense or modality. By contrast, the sentence-final form *taka-i* and the attributive form *taka-(k)i* appear to lack an inflectional ending just like the conjunctive form *taka-ku*, yet unlike *taka-ku*, they indicate non-past tense. The question, then, is how this non-past meaning is formally expressed. Traditionally, the final *-i* in *taka-i* and *taka-(k)i* has often been interpreted as the element that bears the temporal value of "non-past". This interpretation is difficult to justify, however, either from a historical perspective or from the perspective of a descriptive analysis of modern Japanese.

Historically, adjectives originally lacked a non-past vs. past tense opposition and, accordingly, they contained no formal element that encoded tense. For example, in Old Japanese of the 8th century, forms such as *taka-si*, *taka-ki* and *taka-ku* consisted, from the standpoint of the present analysis, solely of the stem, lacking any inflectional ending. However, probably due to the increasing need to mark "past" more explicitly, the past tense suffix *-ki*, which was compatible with verbal stems, and the stem *ar-* of the light verb *ar-i* were employed to form the past tense *taka-k-ar-i-ki*. As Okimori (2017: 84) states, "The auxiliary conjugation known as *kari*-conjugation was formed by the conjunctive form being followed by the irregular verb *ari*, known as a *rahen* verb, through a development such as *ku-ari* > *kari*. This made it possible for the form to attach to a wide range of particles and auxiliary verbs" (translated from Japanese).

This suggests that the sentence-final form *taka-si* and the attributive form *taka-ki*, which originally lacked any overt tense marker, came to express "non-past" through their formal opposition to the newly formed past tense *taka-k-ari-ki* — that is, by lacking the overt tense suffix *-ki*. In other words, from the perspective of formal opposition, the sentence-final and attributive forms came to be analysed as bearing a zero inflectional suffix *-0* specified as [–PAST], which came to stand in opposition to the past-tense inflectional suffix *-ki* specified as [+PAST]. This is parallel to the case of English *cat-0*, where the absence of an overt plural marker in opposition to *cat-s* functions to denote the singular. Based on the above, the morphological structure of the classical forms *taka-si*, *taka-ki*, *taka-ku* and *taka-k-ar-i-ki* may be analysed as in (20):

(20)

(Complex) stem			Ending
VA stem		Light verb stem	Inflexional suffix
VA radical	VA hematic suffix		
<i>taka-</i>	<i>-si</i>	—	<i>-0</i> [−PAST]
<i>taka-</i>	<i>-ki</i>	—	<i>-0</i> [−PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-k</i>	<i>ar-i-</i>	<i>-ki</i> [+PAST]

In the case of modern Japanese as well, it may be assumed that the partial system in (20) remains unchanged, with the past tense suffix *-ki* simply having been replaced by *-ta*, as shown in (21):

(21)

(Complex) stem			Ending
VA stem		Light verb stem	Inflexional suffix
VA radical	VA hematic suffix		
<i>taka-</i>	<i>-i</i>	—	<i>-0</i> [−PAST]
<i>taka-</i>	<i>-(k)i</i>	—	<i>-0</i> [−PAST]
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-ta</i> [+PAST]

What remains to be discussed are the relationship between the sentence-final form *taka-i* and the attributive form *taka-(k)i*, and the conditions that determine their distribution in relation to the conditional form *taka-ke* and the conjunctive form *taka-ku*. The first two forms, *taka-i* and *taka-(k)i*, share the VA radical *taka-* and a zero tense suffix *-0* specified as [−PAST], but differ in the shape of the VA thematic suffix: *-i* in the sentence-final form and *-(k)i* in the attributive form. Correspondingly, they occur in distinct syntactic environments:

(22) Kanozyo ga utsukusi-i:

[_{TP} [_{AP} *kanozyo utsukusi-i*] *-0*] → [_{TP} *kanozyo_i-ga* [_{AP} *t_i t_a*] *utsukusi-i_a-0*]

(23) Utsukusi-(k)i hito:

[_{NP} [_{TP} [_{AP} *OP_i utsukusi-(k)i*] *-0*] *hito_i*] → [_{NP} [_{TP} *OP_i* [_{AP} *t_i t_a*] *utsukusi-(k)i_a-0*] *hito_i*]

As shown in (22), when an AP headed by T *-0* [−past] functions as the complement of TP and that TP is embedded within an NP, the VA thematic suffix appears in the form *-(k)i*. By contrast, when an AP headed likewise by T *-0* [−past] functions as the complement of TP but the TP is not embedded within an NP, as in (23), the VA thematic suffix appears in the form *-i*. As for the remaining forms *-ke* and *-ku*, *-ke* in modern Japanese is exclusively distributed in the environment preceding the suffix *-reba*, while *-ku* is the form taken by the VA thematic suffix in all other syntactic environments.

4. Analysis summary — Conclusion

(i) The morphological structure of the inflected forms of verbal adjectives is as shown in (24):

(24)

(Complex) stem			Ending
VA stem		Light verb stem	Inflexional suffix(es)
VA radical	VA thematic suffix		
<i>taka-</i>	<i>-i</i> ¹	—	<i>-o</i> [–PAST]
<i>taka-</i>	<i>-(k)i</i>	—	<i>-o</i> [–PAST]
<i>taka-</i>	<i>-ke</i>	—	<i>-reba</i>
<i>taka-</i>	<i>-ku</i>	—	— [0 PAST]
<i>taka-</i>	<i>-ku</i>	—	<i>-te</i> [0 PAST]
<i>taka-</i>	<i>-ku</i>	—	<i>-temo</i>
<i>taka-</i>	<i>-ku</i>	—	<i>-tatte</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-ta</i> [+PAST]
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-tar-ō</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-tar-a(ba)</i>
<i>taka-</i>	<i>-k</i>	<i>at-</i>	<i>-tar-i</i>
<i>taka-</i>	<i>-k</i>	<i>ar-</i>	<i>-ō</i>
<i>taka-</i>	<i>-k</i>	<i>ar-</i>	<i>-e</i>

(ii) The VA radical is the element that encodes the lexical meaning specific to each verbal adjective. It can appear independently as a constituent of derived or compound words — for example, *taka-* in *taka-sa* "height", *taka-mi* "loftiness", *taka-nozom-i* "high hope" and *taka-dai* "high platform" — but it cannot, on its own, be used as a syntactic constituent belonging to the category VA at the sentential level — for example, **Kare wa se ga taka* "He is tall"; **Asoko ni se no taka otoko ga iru* "There is a tall man over there"; **Koko kara taka sobieru tō ga mieru* "From here, a tower rising high can be seen"; **Se wa taka-reba ii to yū wake de wa nai* "It is not always the case that it is good if the height is tall".

(iii) The function of the VA thematic suffix *-i* $\approx$ *-(k)i* $\approx$ *-ku* $\approx$ *-ke* is to license the VA radical as a syntactic constituent of category VA within sentential phrase structure. As noted in (ii), *taka-* on its own cannot be used as a VA in a sentence. Only when it combines with *-i*, *-(k)i*, *-ku* or *-ke* — forming *taka-i*, *taka-(k)i*, *taka-ku* or *taka-ke*, for example — does it become syntactically eligible to function as a constituent of category VA — e.g. *Kare wa se ga taka-i*; *Asoko ni se no taka-i otoko ga iru*; *Koko kara taka-ku sobieru tō ga mieru*; **Se wa taka-ke-reba ii to yū wake de wa nai*.

¹ Given that *yo-si*, the classical form of the adjective *yo-i*, *i-i* "to be good", survives in fossilised form in modern Japanese, it is also possible to assume *-si* as the underlying form, with *s* being deleted on the surface in all but exceptional cases.

(iv) The form in which the VA thematic suffix appears — *-i*, *-(k)i*, *-ku* or *-ke* — is determined by the syntactic environment in which the VA stem occurs:

- (25) When a VAP headed by a VA stem serves as the complement of a TP headed by the inflexional suffix *-reba*: → ***-ke***
- (26) When a VAP headed by a VA stem serves as the complement of a TP headed by *-o* [–past], and that TP is embedded within an NP: → ***-(k)i***
- (27) When a VAP headed by a VA stem serves as the complement of a TP headed by *-o* [–past], and that TP is not embedded within an NP: → ***-i***
- (28) In all other cases: → ***-ku***

(v) At the surface level, *-ku* exhibits the phonological alternation $[-kuu] \approx [-k]$, with vowel deletion occurring only when the VA stem combines with the stem *ar-* of the light verb *ar-u*, resulting in the form $[-k]$.

(vi) When these are compared with the inflected forms of the verb *k-u-ru* "to come", which correspond to them semantically, as shown in (29), a strong parallelism in morphological structure can be observed:

(29)

V stem		Ending
V radical	V thematic suffix	Inflexional suffix(es)
<i>k-</i>	<i>-u</i>	<i>-ru</i> [–PAST]
<i>k-</i>	<i>-u</i>	<i>-ru</i> [–PAST]
<i>k-</i>	<i>-u</i>	<i>-reba</i>
<i>k-</i>	<i>-i</i>	— [0 PAST]
<i>k-</i>	<i>-i</i>	<i>-te</i> [0 PAST]
<i>k-</i>	<i>-i</i>	<i>-temo</i>
<i>k-</i>	<i>-i</i>	<i>-tatte</i>
<i>k-</i>	<i>-i</i>	<i>-ta</i> [+PAST]
<i>k-</i>	<i>-i</i>	<i>-tar-ō</i>
<i>k-</i>	<i>-i</i>	<i>-tar-a(ba)</i>
<i>k-</i>	<i>-i</i>	<i>-tar-i</i>
<i>k-</i>	<i>-o</i>	<i>-jō</i>
<i>k-</i>	<i>-o</i>	<i>-i</i>

In vowel-stem verbs, the stem-final vowel can be understood as functioning as the thematic suffix.

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